A High-Performance Distributed System for Multi-Dimensional Flight Delay Correlation & Predictive Intelligence

1. Team Information

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2. Project Motivation / Inspiration

Flight delays cost the U.S. aviation industry over $33 billion annually and affect millions of passengers each year. What makes this problem particularly complex is the "Ripple Effect": a single delayed flight at a major hub can cascade into dozens of downstream disruptions across the national airspace, yet most existing tools treat each delay as an isolated event.

Our team was inspired by the gap between the sheer volume of publicly available aviation data — millions of flight records, continuous meteorological feeds, and FAA infrastructure logs — and the lack of systems that synthesize these sources to explain why delays propagate the way they do. We aim to build a system that not only identifies delay patterns but quantifies their root causes and predicts future disruptions before they occur.

3. Project Description

SkyPath Analytics is a distributed data pipeline and predictive intelligence platform designed to integrate multi-source aviation datasets and deliver scalable flight delay modeling, analysis, and forecasting.

The system is organized into four tightly coupled layers. At the foundation, the Data Acquisition and Storage Layer ingests and harmonizes data from three primary sources: BTS Historical Flight Records covering more than six million flights per year, IEM/NOAA Meteorological Data in METAR and TAF formats, and FAA Airport Infrastructure Metadata. All ingested data is persisted and schema-enforced within a Hadoop HDFS-based Data Lake, establishing a unified source of truth that guarantees consistency for all downstream processing.

Built directly on top of this foundation, the Processing Layer serves as the computational engine of the system, leveraging Apache Spark's distributed in-memory architecture to execute large-scale ETL workflows and feature engineering. This layer employs Broadcast Joins to temporally align flight records with precise weather snapshots, applies Timezone Normalization to ensure cross-regional temporal consistency across the national airspace, and utilizes Advanced Window Functions to perform Tail Number Tracking — a process that reconstructs each aircraft's full daily flight chain in order to quantify cumulative delay offsets and model inter-flight propagation patterns.

The enriched feature set produced by the processing layer is then consumed by the Analytics and Machine Learning Intelligence Layer, which transforms structured data into actionable predictive insights. Multivariate Regression is applied to decouple delay causation, isolating carrier-driven, weather-driven, and NAS-driven contributions within a unified attribution framework. Complementing this, Propagation Modeling traces how disruptions originating at major hub airports ripple outward through the broader network. Leveraging Spark MLlib, the layer trains both Gradient

Boosted Tree and Random Forest models to generate real-time delay probability scores and estimated arrival offset predictions for scheduled departures.

Finally, the Visualization Layer translates these analytical outputs into an interactive, stakeholder-facing decision-support tool. Built on React.js and powered by ECharts and Mapbox GL JS, the dashboard delivers a high-fidelity geospatial interface that overlays flight trajectory data with live weather context. Users can dynamically filter performance metrics by airline, airport, or time-of-day, and leverage a dedicated predictive Risk Lookup utility to support proactive scheduling and airspace management decisions.

4. Expected Outcomes / Deliverables

By the end of the project, the team expects to produce six core deliverables.
The first is a Unified Data Pipeline — a fully operational ETL pipeline that ingests, cleans, and joins flight records, meteorological observations, and airport infrastructure data, with all outputs stored and queryable on HDFS. Building on this foundation, the team will publish a Ripple Effect Feature Dataset, a processed analytical dataset that captures aircraft-level delay propagation chains, causal factor breakdowns, and hub disruption scores derived from the Tail Number Tracking methodology. On the machine learning side, the project will produce a set of Trained Predictive Models, specifically Gradient Boosted Tree and Random Forest models trained via Spark MLlib, capable of predicting both flight delay probability and estimated delay magnitude for scheduled departures. Model performance will be fully documented through cross-validation metrics including RMSE and F1-score. To make these models accessible to the frontend, the team will also develop a REST API for Inference — a lightweight serving layer that exposes real-time delay predictions for integration with the dashboard.

The primary user-facing deliverable is an Interactive Dashboard, a deployed web application that provides geospatial delay visualization, airline and airport performance analytics, and a dedicated Risk Lookup utility enabling proactive, data-driven scheduling decisions. Finally, the project will conclude with a comprehensive Project Report documenting all data sources, engineering methodology, model performance outcomes, key architectural decisions, and analytical findings across the full system.

5. Dataset / Data Sources

SkyPath Analytics draws on three primary datasets and two optional supplementary sources, all of which are publicly accessible.

The analytical backbone of the pipeline is the BTS On-Time Performance Records, sourced from the U.S. Bureau of Transportation Statistics. This database tracks domestic flight operations across all major U.S. carriers and provides field-selectable CSV downloads at the official BTS portal (https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr), with full field definitions available at https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FGJ and a pre-cleaned Kaggle mirror at https://www.kaggle.com/datasets/daryaheyko/airline-on-time-statistics-and-delay-causes-bts. Key fields include flight date, origin and destination airport, carrier code, scheduled and actual departure and arrival times, delay minutes broken down by cause — covering carrier, weather, NAS, security, and late aircraft — as well as tail number and cancellation or diversion flags. At over six million records per year, multi-year historical pulls from this source will form the primary input to the ETL pipeline.

Meteorological context is provided by the IEM ASOS/METAR Dataset, sourced from the Iowa Environmental Mesonet, which aggregates hourly and sub-hourly surface weather observations from more than 900 U.S. airport weather stations. Data can be accessed through the ASOS download interface at https://mesonet.agron.iastate.edu/request/download.phtml, with dataset documentation available at https://mesonet.agron.iastate.edu/info/datasets/metar.html. The key fields extracted include station ID, observation timestamp, temperature, dewpoint, wind speed and direction, visibility, precipitation, and condition codes such as fog or thunderstorm indicators. Within the Spark pipeline, these weather snapshots will be temporally joined to flight departure records via Broadcast Joins, enabling precise causal attribution of weather-driven delay contributions.

The third primary source is Airport Infrastructure Metadata, which supplies the static reference data required to resolve IATA and ICAO code mappings, geolocations, and airport capacity attributes across the full flight network. Multiple sources are available for this layer, including OurAirports(https://ourairports.com/data/), the DataHub airport codes dataset (https://datahub.io/core/airport-codes), the FAA's official data portal (https://www.faa.gov/data), and the Python airportsdata library (https://pypi.org/project/airportsdata/). Key fields include IATA and ICAO codes, airport name, municipality, latitude and longitude, elevation, and airport type. This dataset serves a dual purpose: enriching the Mapbox geospatial dashboard layer and bridging the code system mismatch between BTS flight records, which use IATA codes, and IEM weather stations, which use ICAO identifiers.

Two supplementary datasets may be incorporated to further enrich the analytical model. The BTS T-100 Segment Data (https://www.transtats.bts.gov/Fields.asp?Table_ID=293) provides route-level passenger volume and available seat miles, enabling the system to weight delay impact by flight capacity and prioritize high-volume disruption corridors in the dashboard. Separately, the U.S. Public Holiday Calendar, accessible via the Nager.Date REST API at ***https://date.nager.at/api/v3/publicholidays/{year}/US***, will be used to encode holiday proximity as a binary or ordinal feature in the machine learning models, capturing the well-documented surge in delay rates during peak travel periods such as Thanksgiving and Independence Day.

6. Justification

SkyPath Analytics qualifies as a Big Data project across multiple dimensions, each of which independently justifies the adoption of a distributed computing framework.

From a volume perspective, the BTS flight dataset alone contributes over six million records per year. When combined with high-frequency METAR weather observations — reported every 30 to 60 minutes across more than 900 U.S. airport stations — and supplementary FAA infrastructure metadata, the total dataset footprint reaches tens of gigabytes to low terabytes. This scale far exceeds the practical capacity of single-machine processing tools such as pandas or scikit-learn, making distributed ingestion and computation not a design preference but an operational necessity.

The project also exhibits significant data variety. Rather than operating on a single homogeneous source, the pipeline must reconcile structured tabular records from BTS flight logs, semi-structured time-series sensor data from METAR and TAF meteorological feeds, and relational metadata schemas from FAA airport capacity tables. Handling this heterogeneity — across differing formats, temporal granularities, and join semantics — requires a unified pipeline architecture capable of normalizing and synthesizing fundamentally different data representations.

On the velocity dimension, weather conditions and flight statuses are inherently dynamic, changing on a continuous basis. While the core analytical pipeline operates in batch mode over historical data, the system architecture is explicitly designed to support near-real-time inference serving through the REST API layer, which in turn requires efficient incremental feature computation to deliver low-latency predictions for scheduled departures.

Finally, and most critically, the project demands distributed processing as a technical requirement rather than a stylistic choice. Core analytical operations — including Tail Number Tracking implemented via Spark Window Functions across millions of flight legs, large-scale Broadcast Joins that temporally align flight records with granular weather snapshots, and distributed model training through Spark MLlib — are computationally infeasible on a single machine within any practical time constraint. Apache Spark is therefore the appropriate and necessary foundation for this workload, selected on the basis of technical fit rather than convenience.

Implementation Roadmap:

Phase	Duration	Focus
Phase I	Weeks 1	Data ingestion, schema optimization, and HDFS/S3 environment setup.
Phase II	Weeks 2-3	ETL pipeline development and "Ripple Effect" feature logic implementation.
Phase III	Weeks 4-5	ML model training, cross-validation, and REST API development for inference.
Phase IV	Weeks 6-7	Frontend dashboard integration, UI/UX polishing, and final UAT (User Acceptance Testing).